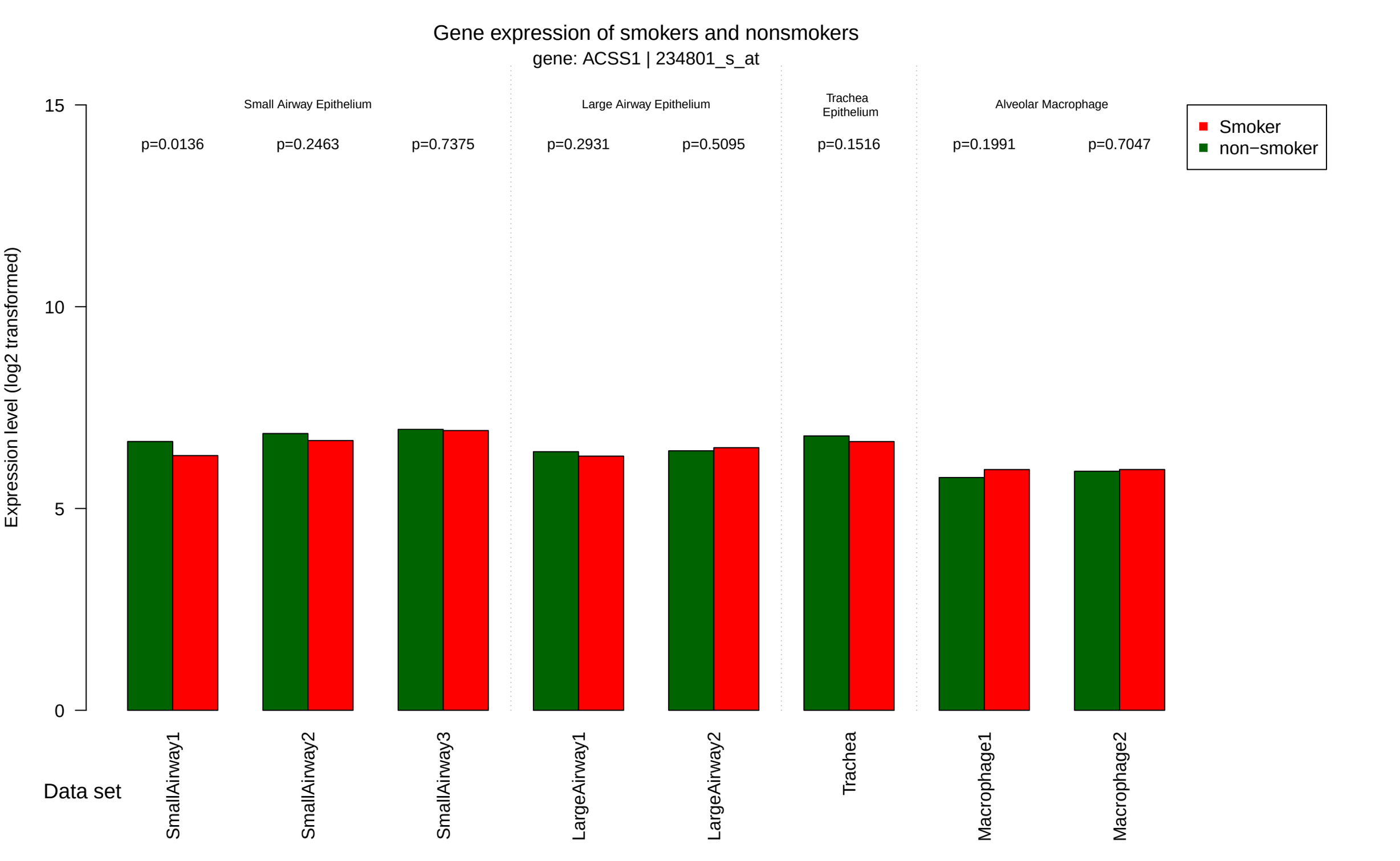
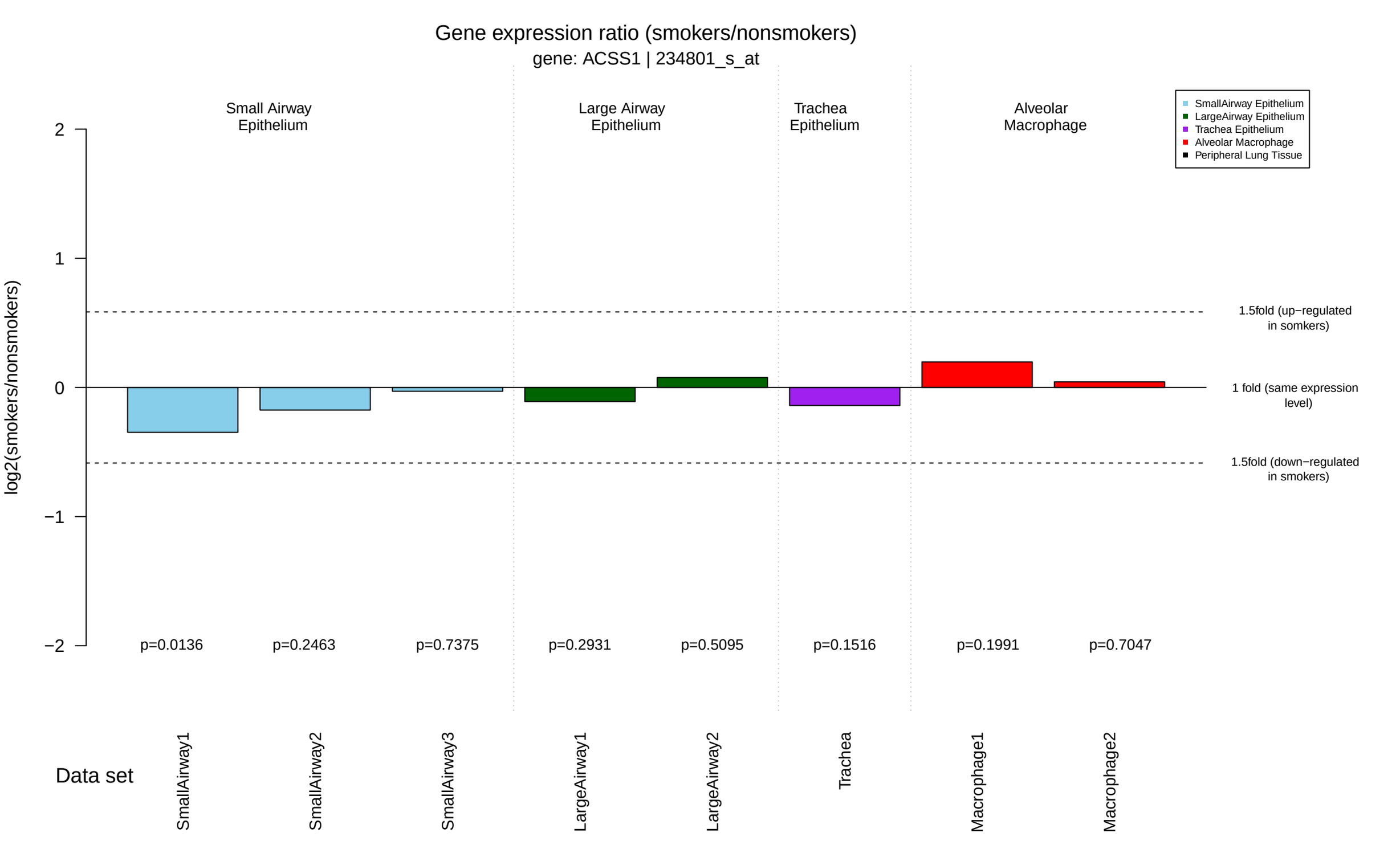
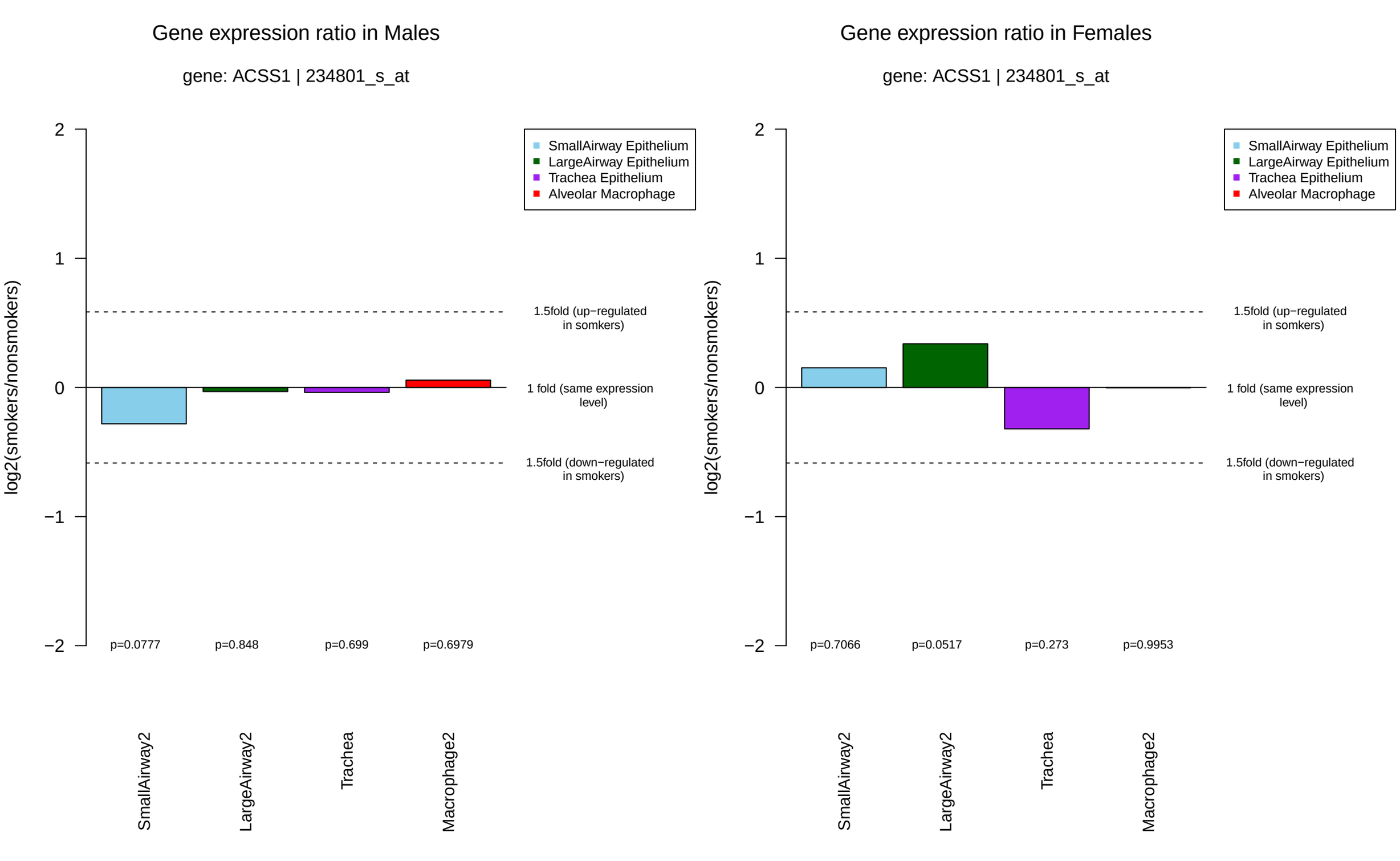
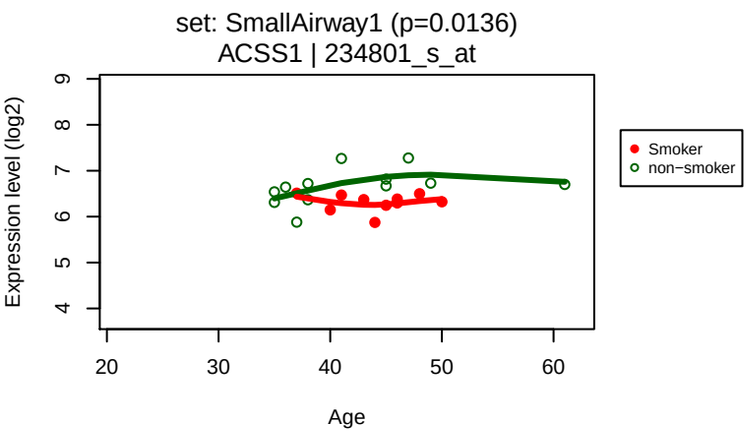
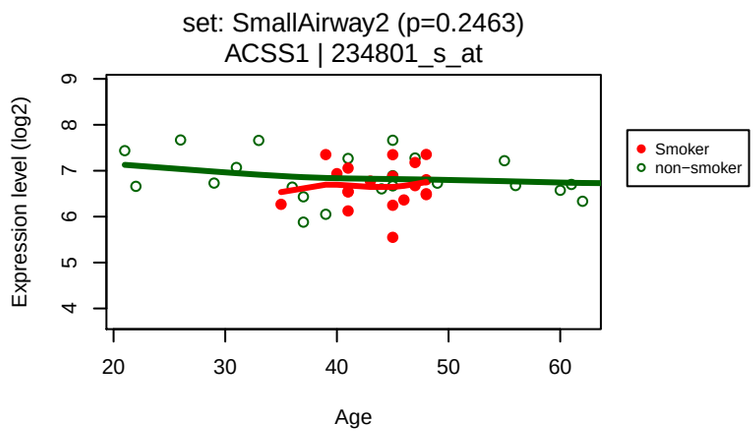
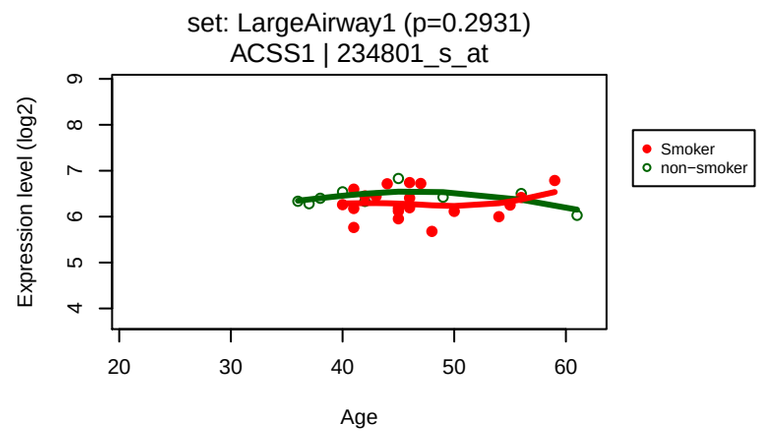

Gene expression of smokers and nonsmokers
gene: ACSS1 | 234801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0136
p=0.2463
p=0.7375
p=0.2931
p=0.5095
p=0.1516
p=0.1991
p=0.7047
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

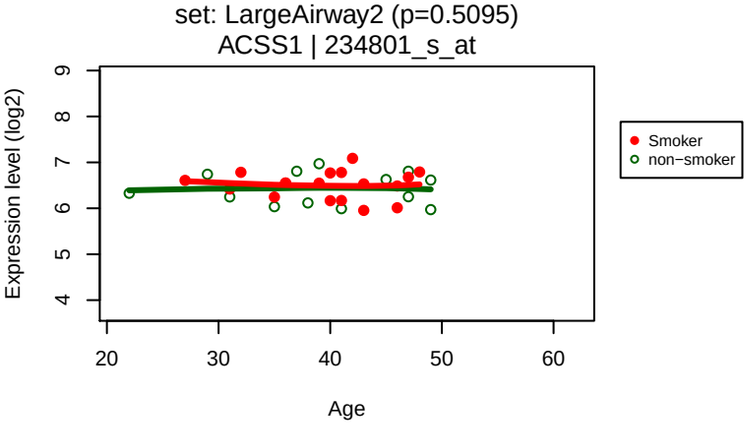

Gene expression ratio (smokers/nonsmokers)
gene: ACSS1 | 234801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0136
p=0.2463
p=0.7375
p=0.2931
p=0.5095
p=0.1516
p=0.1991
p=0.7047
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

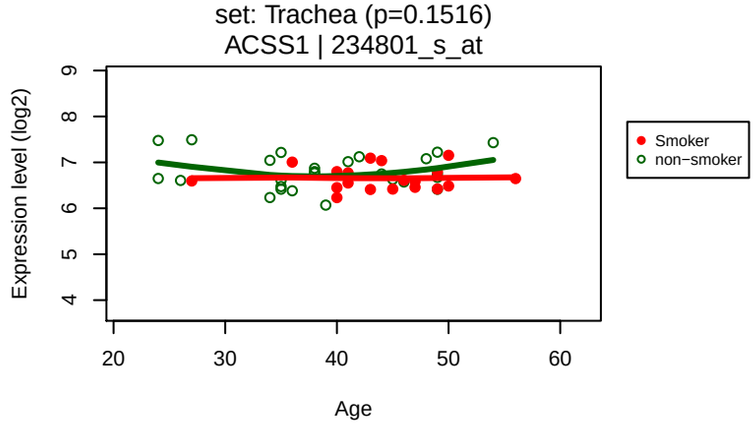

Gene expression ratio in Males
gene: ACSS1 | 234801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0777
p=0.848
p=0.699
p=0.6979
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSS1 | 234801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7066
p=0.0517
p=0.273
p=0.9953
SmallAirway2
LargeAirway2
Trachea
Macrophage2

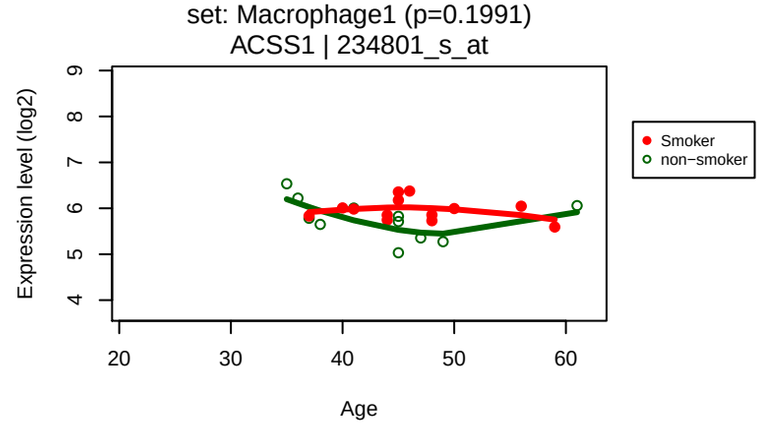

set: SmallAirway1 (p=0.0136)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker

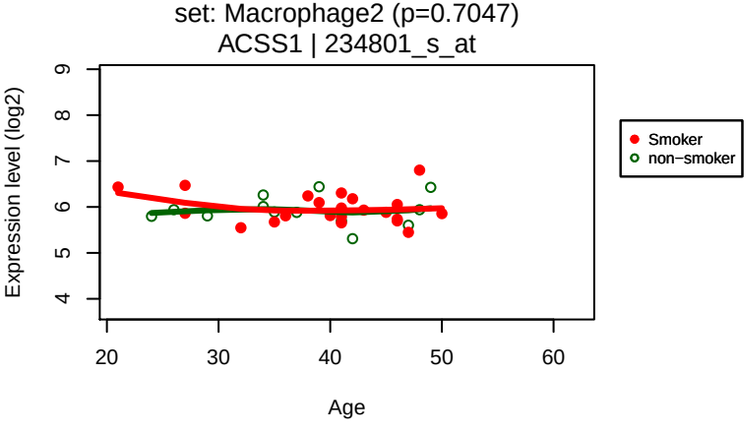

set: SmallAirway2 (p=0.2463)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker

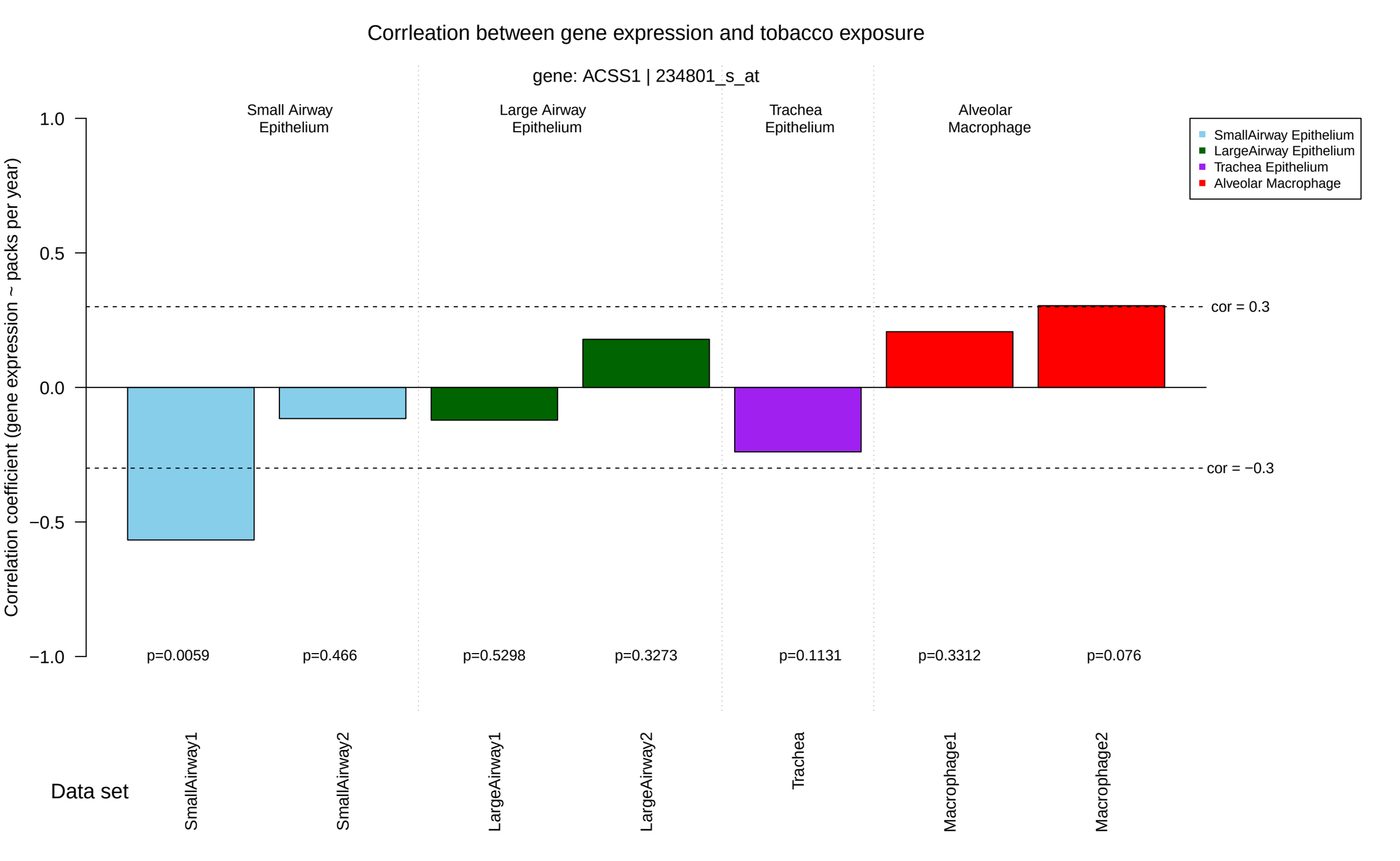

set: LargeAirway1 (p=0.2931)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5095)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1516)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1991)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7047)
ACSS1 | 234801_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSS1 | 234801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0059
p=0.466
p=0.5298
p=0.3273
p=0.1131
p=0.3312
p=0.076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2